Emergence and Prediction of New Financial Risk Categories

A Text-Based Analysis of News and Regulatory Disclosures

Executive Summary

This project investigates how new financial risk categories emerge and become material for firms and markets. Using a decade of major U.S. financial news combined with firm-level regulatory disclosures from 10-K and 8-K filings, the study reconstructs historical episodes of risk emergence and identifies measurable early indicators. Building on these retrospective findings, the project develops a forward-looking framework designed to estimate the probability that a candidate theme (such as AI-related risk) will evolve into an institutionalized and widely disclosed financial risk category.

The project contributes to the literature by providing a reproducible and data-driven definition of risk emergence, integrating textual signals from multiple information channels, and translating qualitative narratives into quantitative indicators suitable for econometric analysis and early-warning systems.

1 Research Questions and Objectives

The project is structured around two primary research questions.

RQ1 (Retrospective). How does a new financial risk category emerge? Which observable indicators in the information environment precede its broad recognition and institutionalization in corporate disclosures?

RQ2 (Forward-looking). Given information available up to the present, how can the emergence of a new risk category be anticipated, and how can the probability of such emergence be estimated for candidate themes such as AI-related risk?

The main objectives are:

- to define an operational and reproducible notion of financial risk emergence;
- to construct quantitative indicators capturing attention, novelty, diffusion, and structural integration;
- to estimate typical lead–lag relationships between news, 8-K disclosures, and 10-K risk-factor adoption;
- to develop and validate an early-warning framework for emerging risks;
- to produce research outputs suitable for academic publication and practical risk monitoring.

2 Data Sources and Corpus Construction

The empirical analysis combines four complementary information channels:

- **Financial news:** full-text articles from major U.S. newspapers with a financial focus, including publication dates and metadata.
- **10-K filings:** Item 1A (Risk Factors) extracted on a firm-year basis.
- **8-K filings:** event-driven disclosures capturing material incidents and emerging concerns.
- **Regulatory and normative texts (optional):** laws, regulations, supervisory guidance, consultation papers, and standards issued by regulatory authorities and standard-setting bodies. This channel is treated as optional and market-dependent, reflecting heterogeneity in the availability and maturity of normative disclosure across jurisdictions.

All documents are time-stamped and organized into a unified corpus. Preprocessing includes boilerplate removal, deduplication, paragraph segmentation, and normalization of firm identifiers and named entities.

3 Defining Financial Risk Emergence

A central methodological requirement is a definition of risk emergence that is measurable and replicable. A candidate risk is said to *emerge* at time t^* when multiple indicators jointly transition from low background levels to a persistent high state, indicating diffusion and institutionalization rather than isolated mentions.

Operational criteria include:

- a sustained increase in news attention;
- semantic consolidation of the underlying concept;
- a persistent rise in firm-level disclosure;
- structural integration with, and eventual differentiation from, existing risk categories.
- where available, progressive formalization in regulatory or normative texts, such as the introduction of definitions, reporting obligations, or supervisory expectations.

4 Indicators for Retrospective Analysis

4.1 News-Based Indicators

News-based indicators capture the informational environment in which new risks first appear:

- coverage intensity and burst dynamics;
- novelty and semantic drift of risk-related language;
- expansion of entity co-mention networks;
- diversification of contexts across industries and topics.

4.2 Disclosure-Based Indicators

Indicators derived from 10-K and 8-K filings capture institutional adoption:

- diffusion rate across firms;
- disclosure intensity within filings;
- co-occurrence patterns with other risk categories;
- lead–lag relationships between 8-K and 10-K disclosures.

4.3 Regulatory and Normative Indicators (Optional)

When available, regulatory and normative texts provide an additional layer of information on risk emergence. Indicators derived from this channel include:

- frequency and timing of references to a candidate risk in laws, regulations, supervisory statements, and consultation documents;
- degree of formalization, measured by the transition from descriptive mentions to explicit definitions, principles, or reporting requirements;
- cross-jurisdictional diffusion, capturing whether a risk appears sequentially or simultaneously across different regulatory authorities;
- alignment between regulatory language and corporate disclosure terminology, indicating convergence toward a shared conceptual framework.

5 Methodological Framework

The empirical framework proceeds in four steps.

5.1 Candidate Theme Identification

Candidate themes are identified in news data using time-aware topic discovery and burst detection. Themes displaying sustained growth and expanding contextual usage are retained as potential emerging risks.

5.2 Theme Tagging and Measurement

Identified themes are mapped to news and disclosure texts using supervised or weakly supervised classification. Document-level scores are aggregated into time series capturing attention, diffusion, and co-occurrence dynamics.

5.3 Estimating Emergence Mechanisms

Two complementary strategies are employed:

- **Event-history (hazard) models**, where the event of interest is the first-time disclosure of a risk by a firm;

- **Change-point detection**, applied to diffusion and intensity series to identify structural breaks.

5.4 Validation

Validation relies on out-of-sample prediction, robustness checks across alternative specifications, and qualitative alignment with documented historical events. Market-based validation may additionally assess reactions in returns or volatility.

6 Forward-Looking Early Warning Framework

The forward-looking component estimates the probability that a candidate theme will transition into an institutionalized risk category within a given horizon (e.g., 3, 6, or 12 months). The framework uses only information available up to the forecast date and is evaluated in a rolling-window setting.

A key application concerns AI-related risk, monitored through sub-themes such as model risk, regulation, data governance, security vulnerabilities, intellectual property, and workforce impact.

7 Expected Contributions

The project delivers:

- a reproducible definition of financial risk emergence;
- a measurement toolkit for textual risk indicators;
- empirical evidence on lead–lag dynamics between news and disclosures;
- an early-warning model for emerging risk categories;
- a research-ready dataset for further econometric analysis.

A Appendix A: Glossary of Technical Terms

Burst detection (Kleinberg algorithm). A method to identify time intervals in which the frequency of events increases sharply relative to a baseline, signaling abnormal attention.

Semantic drift. The evolution of a concept's meaning over time, measured through changes in semantic representations.

Novelty. A measure of how different current usage of a concept is relative to its historical baseline.

Disclosure diffusion. The spread of a risk category across firms, typically measured as the fraction of firms disclosing the risk.

Co-occurrence. The joint appearance of multiple risks within the same document or textual unit.

Change-point detection. Statistical methods that identify structural changes in time series.

Event-history (hazard) modeling. A framework for modeling the timing of events, such as first-time risk disclosure.

Rolling-window evaluation. A forecasting evaluation procedure that uses only past information at each forecast date.

B Appendix B: Identification of the Risk Emergence Time t^*

For each candidate risk, a disclosure diffusion curve is constructed as:

$$D_r(t) = \frac{\text{Number of firms disclosing risk } r \text{ at time } t}{\text{Total number of firms at time } t}.$$

The emergence time t^* is defined as the point at which this series exhibits a statistically significant structural change. Prior to t^*, diffusion fluctuates around a low baseline, reflecting sporadic mentions. After t^*, diffusion increases persistently, indicating institutionalization.

Change-point detection methods are applied to estimate t^* in a data-driven manner. The resulting estimate is subsequently assessed for plausibility by comparison with external historical timelines, such as major incidents, regulatory changes, or technological shifts. This validation step supports interpretation without imposing external information on the estimation procedure itself.

Identifying t^* provides both an operational definition of risk emergence and a reference point for analyzing lead–lag relationships between news coverage, event-driven disclosures, and annual risk-factor reporting.